

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)

11
priors compared

32,572
held-out plate appearances

3.03%
held-out HR rate

0.6007
best AUC · KNN comps

served prior

Held-out scores

Fitted on plate appearances before 2026-08-28, scored on the 32,572 that came after (actual HR rate 3.03%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5853	1.56×	0.02935	0.13493	0.00452
KNN comps Prior is the pooled rate of the nearest comparables.	0.6007	1.51×	0.02932	0.13397	0.00417
SVR Support-vector regression on the same feature matrix.	0.5980	1.50×	0.02935	0.13448	0.00604
Random forest Forest regression; a learned similarity metric.	0.5972	1.45×	0.02935	0.13433	0.00596
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6007	1.51×	0.02932	0.13397	0.00417
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6007	1.51×	0.02932	0.13397	0.00417
Linear regression Ordinary least squares on the same features.	0.5969	1.50×	0.02933	0.13427	0.00421
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5978	1.54×	0.02935	0.13440	0.00622
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5971	1.53×	0.02934	0.13427	0.00486
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5973	1.46×	0.02935	0.13439	0.00611

BLEND WEIGHTS (FOREST VARIANT)

KNN 1.00 SVR 0.00 RF 0.00

SHRINKAGE K PER PRIOR

league 248 knn 248 svr 248 rf 248 core 248 ensemble 248
linear 248 logistic 248 nn 248 xgb 248

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,530 PA, actual HR rate 2.35%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5569	1.45×	0.02299	0.11186	0.01009
KNN comps Prior is the pooled rate of the nearest comparables.	0.5834	1.33×	0.02302	0.11167	0.01228
SVR Support-vector regression on the same feature matrix.	0.5836	1.33×	0.02316	0.11456	0.01754
Random forest Forest regression; a learned similarity metric.	0.5925	1.08×	0.02317	0.11288	0.01708
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5834	1.33×	0.02302	0.11167	0.01228
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5834	1.33×	0.02302	0.11167	0.01228
Linear regression Ordinary least squares on the same features.	0.5861	1.08×	0.02310	0.11311	0.01521
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5910	1.08×	0.02311	0.11262	0.01479
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5854	1.20×	0.02306	0.11227	0.01442
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6004	1.20×	0.02318	0.11289	0.01705

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Munetaka Murakami 14.0%, Randal Grichuk 12.3%, Miguel Vargas 12.2%, Tommy Pham 11.7%, Jo Adell 11.3%, Chase Meidroth 8.7%
Flat league	6/6	Munetaka Murakami 12.4%, Randal Grichuk 11.1%, Jo Adell 10.7%, Miguel Vargas 10.7%, Chase Meidroth 9.8%, Tommy Pham 9.7%
KNN comps	6/6	Munetaka Murakami 13.9%, Randal Grichuk 12.1%, Miguel Vargas 12.1%, Tommy Pham 11.7%, Jo Adell 11.2%, Chase Meidroth 8.8%
SVR	6/6	Munetaka Murakami 14.5%, Randal Grichuk 12.8%, Miguel Vargas 12.1%, Jo Adell 11.6%, Tommy Pham 10.2%, Chase Meidroth 9.1%
Random forest	6/6	Munetaka Murakami 14.2%, Randal Grichuk 12.1%, Miguel Vargas 11.9%, Jo Adell 11.4%, Tommy Pham 10.8%, Chase Meidroth 9.1%
Core (KNN+SVR)	6/6	Munetaka Murakami 13.9%, Randal Grichuk 12.1%, Miguel Vargas 12.1%, Tommy Pham 11.7%, Jo Adell 11.2%, Chase Meidroth 8.8%
Forest blend	6/6	Munetaka Murakami 13.9%, Randal Grichuk 12.1%, Miguel Vargas 12.1%, Tommy Pham 11.7%, Jo Adell 11.2%, Chase Meidroth 8.8%
Linear regression	6/6	Munetaka Murakami 13.8%, Miguel Vargas 12.1%, Randal Grichuk 12.0%, Tommy Pham 11.9%, Jo Adell 11.3%, Chase Meidroth 8.9%
Logistic regression	5/6	Munetaka Murakami 14.0%, Randal Grichuk 12.9%, Miguel Vargas 12.3%, Tommy Pham 11.7%, Jo Adell 11.6%, Colson Montgomery 9.0%
Neural net (torch)	6/6	Munetaka Murakami 14.0%, Randal Grichuk 12.6%, Miguel Vargas 12.2%, Tommy Pham 11.9%, Jo Adell 11.6%, Chase Meidroth 8.7%
XGBoost	6/6	Munetaka Murakami 14.5%, Randal Grichuk 12.3%, Miguel Vargas 12.1%, Jo Adell 11.5%, Tommy Pham 10.1%, Chase Meidroth 9.0%

Hitter by hitter

The top 40 bats by the served probability.

Pitcher Performance Metrics - Q3 2024											
HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET	
1Munetaka Murakami	14.0% on slate	12.4%	13.9%	14.5%	14.2%	13.9%	13.9%	13.8%	14.0%		
2Randal Grichuk	12.3% on slate	11.1%	12.1%	12.8%	12.1%	12.1%	12.1%	12.0%	12.9%		
3Miguel Vargas	12.2% on slate	10.7%	12.1%	12.1%	11.9%	12.1%	12.1%	12.1%	12.3%		
4Tommy Pham	11.7% on slate	9.7%	11.7%	10.2%	10.8%	11.7%	11.7%	11.9%	11.7%		
5Jo Adell	11.3% on slate	10.7%	11.2%	11.6%	11.4%	11.2%	11.2%	11.3%	11.6%		
6Chase Meidroth	8.7% on slate	9.8%	8.8%	9.1%	9.1%	8.8%	8.8%	8.9%	8.8%		
7Nathaniel Lowe	8.3% on slate	7.8%	8.2%	8.0%	8.5%	8.2%	8.2%	8.0%	8.0%		
8Colson Montgomery	8.2% on slate	7.2%	8.2%	7.9%	8.2%	8.2%	8.2%	8.5%	9.0%		
9Chase DeLauter	8.1% on slate	8.2%	8.0%	7.9%	7.9%	8.0%	8.0%	7.9%	7.8%		
10Kyle Teel	7.9% on slate	7.5%	8.0%	7.4%	7.8%	8.0%	8.0%	7.8%	7.1%		
11Angel Martínez	7.8% on slate	8.0%	7.8%	7.9%	7.8%	7.8%	7.8%	7.8%	7.8%		
12Travis Bazzana	6.9% on slate	6.9%	6.7%	6.5%	7.0%	6.7%	6.7%	6.7%	6.5%		
13Brenton Doyle	6.8% on slate	8.5%	7.4%	7.3%	7.0%	7.4%	7.4%	7.3%	7.1%		
14José Ramírez	6.5% on slate	6.8%	6.7%	6.2%	6.6%	6.7%	6.7%	6.8%	7.0%		
15Braden Montgomery	5.4% on slate	6.2%	5.1%	4.8%	5.2%	5.1%	5.1%	4.9%	4.8%		
16Austin Hedges	4.9% on slate	6.0%	5.4%	5.3%	5.2%	5.4%	5.4%	5.4%	5.1%		
17Bryan Rocchio	4.8% on slate	5.2%	4.7%	4.9%	4.4%	4.7%	4.7%	4.6%	4.6%		
18Steven Kwan	3.0% on slate	5.6%	3.7%	2.9%	2.3%	3.7%	3.7%	1.8%	3.0%		

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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